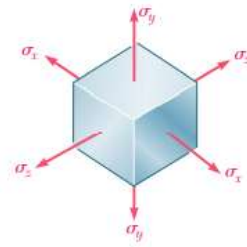


Type equation here.

MULTI-AXIAL LOADING: Generalised Hooke's Law

So far, we have been discussing uni-axial loading. Now, consider an element of the material is subjected to axial normal stresses in three mutually perpendicular directions. Let the side of the cube be a . The normal stresses will change the dimensions of the cube. If the material is stresses within the linear elastic range, we can apply the principle of superposition. That is, the effect of each stress is considered separately and total deformation is the algebraic sum of the individual deformations.



Only normal stresses will cause a change in dimension (or change in volume), shear stress will not. Shear stress will cause only a change in shape.

The strains produced by each normal stress in three mutually perpendicular directions are given in the table below.

	Strain in the x-direction	Strain in the y-direction	Strain in the z-direction
Tensile stress in the x-direction, σ_{xx}	$\frac{\sigma_{xx}}{E}$ (linear strain)	$-\nu \frac{\sigma_{xx}}{E}$ (lateral strain)	$-\nu \frac{\sigma_{xx}}{E}$ (lateral strain)
Tensile stress in the y-direction, σ_{yy}	$-\nu \frac{\sigma_{yy}}{E}$ (lateral strain)	$\frac{\sigma_{yy}}{E}$ (linear strain)	$-\nu \frac{\sigma_{yy}}{E}$ (lateral strain)
Tensile stress in the z-direction, σ_{zz}	$-\nu \frac{\sigma_{zz}}{E}$ (lateral strain)	$-\nu \frac{\sigma_{zz}}{E}$ (lateral strain)	$\frac{\sigma_{zz}}{E}$ (linear strain)

The net normal strain in the x-direction, $\epsilon_{xx} = \frac{\sigma_{xx}}{E} - \nu \frac{\sigma_{yy}}{E} - \nu \frac{\sigma_{zz}}{E}$

The net normal strain in the y-direction, $\epsilon_{yy} = \frac{\sigma_{yy}}{E} - \nu \frac{\sigma_{xx}}{E} - \nu \frac{\sigma_{zz}}{E}$

The net normal strain in the z-direction, $\epsilon_{zz} = \frac{\sigma_{zz}}{E} - \nu \frac{\sigma_{xx}}{E} - \nu \frac{\sigma_{yy}}{E}$

The shear strains

$$\epsilon_{xy} = \frac{\gamma_{xy}}{2} = \frac{\tau_{xy}}{2G} = \frac{\tau_{xy}}{2 \frac{E}{2(1+\nu)}} = \frac{\tau_{xy}(1+\nu)}{E} \quad \ll \gg \quad \epsilon_{yz} = \frac{\tau_{yz}(1+\nu)}{E} \quad \ll \gg \quad \epsilon_{zx} = \frac{\tau_{zx}(1+\nu)}{E}$$

This is known as **generalised Hooke's Law** for multi-axial loading.

$$\begin{bmatrix} \epsilon_{xx} \\ \epsilon_{yy} \\ \epsilon_{zz} \\ \epsilon_{xy} \\ \epsilon_{yz} \\ \epsilon_{zx} \end{bmatrix} = \frac{1}{E} \begin{bmatrix} 1 & -\nu & -\nu & 0 & 0 & 0 \\ -\nu & 1 & -\nu & 0 & 0 & 0 \\ -\nu & -\nu & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1+\nu & 0 & 0 \\ 0 & 0 & 0 & 0 & 1+\nu & 0 \\ 0 & 0 & 0 & 0 & 0 & 1+\nu \end{bmatrix} \begin{bmatrix} \sigma_x \\ \sigma_y \\ \sigma_z \\ \tau_{xy} \\ \tau_{yz} \\ \tau_{zx} \end{bmatrix}$$

It is left as an exercise to the students to express the stresses in terms of the strain values as shown below.

$$\sigma_{xx} = \frac{\nu E}{(1 + \nu)(1 - 2\nu)} (\varepsilon_{xx} + \varepsilon_{yy} + \varepsilon_{zz}) + \frac{E}{1 + \nu} \varepsilon_{xx} = \lambda e + 2G\varepsilon_{xx}$$

$$\begin{Bmatrix} \sigma_x \\ \sigma_y \\ \sigma_z \\ \tau_{xy} \\ \tau_{yz} \\ \tau_{zx} \end{Bmatrix} = \frac{E}{(1 + \nu)(1 - 2\nu)} \begin{bmatrix} 1 - \nu & \nu & \nu & 0 & 0 & 0 \\ \nu & 1 - \nu & \nu & 0 & 0 & 0 \\ \nu & \nu & 1 - \nu & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 - 2\nu & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 - 2\nu & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 - 2\nu \end{bmatrix} \begin{Bmatrix} \varepsilon_{xx} \\ \varepsilon_{yy} \\ \varepsilon_{zz} \\ \varepsilon_{xy} \\ \varepsilon_{yz} \\ \varepsilon_{xz} \end{Bmatrix}$$

Due to the normal strains in the three mutually perpendicular directions, the dimensions of the cube (a, a, a) will change to $(a + a\varepsilon_{xx}, a + a\varepsilon_{yy}, a + a\varepsilon_{zz})$.

The initial volume of the cube, $V = a^3$

The final volume of the cube,

$$\begin{aligned} V' &= (a + a\varepsilon_{xx})(a + a\varepsilon_{yy})(a + a\varepsilon_{zz}) = a^3(1 + \varepsilon_{xx})(1 + \varepsilon_{yy})(1 + \varepsilon_{zz}) \\ &= a^3\{1 + (\varepsilon_{xx} + \varepsilon_{yy} + \varepsilon_{zz}) + (\varepsilon_{xx}\varepsilon_{yy} + \varepsilon_{yy}\varepsilon_{zz} + \varepsilon_{xx}\varepsilon_{zz}) + \varepsilon_{xx}\varepsilon_{yy}\varepsilon_{zz}\} \end{aligned}$$

Since the strain values are very small, their products are still smaller and hence they can be neglected.

$$V' = a^3\{1 + (\varepsilon_{xx} + \varepsilon_{yy} + \varepsilon_{zz})\}$$

The change in volume, $\Delta V = V' - V = a^3(\varepsilon_{xx} + \varepsilon_{yy} + \varepsilon_{zz})$

$$\text{The volumetric strain (Dilatation, } e) = \frac{\text{Change in volume}}{\text{Original volume}} = \frac{\Delta V}{V} = (\varepsilon_{xx} + \varepsilon_{yy} + \varepsilon_{zz})$$

$$e = \left(\frac{\sigma_{xx}}{E} - \nu\frac{\sigma_{yy}}{E} - \nu\frac{\sigma_{zz}}{E}\right) + \left(\frac{\sigma_{yy}}{E} - \nu\frac{\sigma_{xx}}{E} - \nu\frac{\sigma_{zz}}{E}\right) + \left(\frac{\sigma_{zz}}{E} - \nu\frac{\sigma_{xx}}{E} - \nu\frac{\sigma_{yy}}{E}\right)$$

$$e = \frac{\sigma_{xx} + \sigma_{yy} + \sigma_{zz}}{E} - \frac{2\nu}{E}(\sigma_{xx} + \sigma_{yy} + \sigma_{zz}) = (\sigma_{xx} + \sigma_{yy} + \sigma_{zz})\left(\frac{1 - 2\nu}{E}\right)$$

If $\sigma_{xx} = \sigma_{yy} = \sigma_{zz} = \sigma$ (Spherical stress or Hydrostatic stress)

$$e = 3\sigma\left(\frac{1 - 2\nu}{E}\right)$$

$$\frac{\sigma}{e} = \frac{E}{3(1 - 2\nu)}$$

The ratio of spherical stress to volumetric strain(dilatation) is known as Bulk Modulus (K)

$$K = \frac{E}{3(1 - 2\nu)}$$

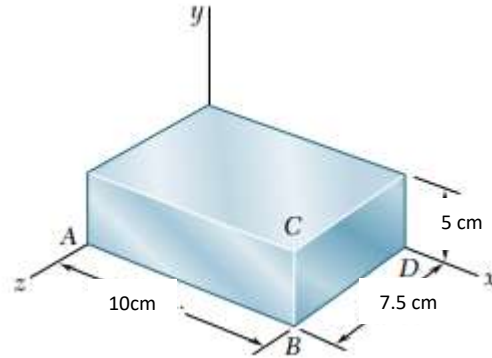
A positive σ (tensile stress) will produce a positive dilatation (increase in volume) only if

$$(1 - 2\nu) > 0 \quad \Leftrightarrow \quad \nu < \frac{1}{2}$$

For any engineering material, the Poisson's ratio varies between 0 and 0.5.

Example 1

The steel block shown is subjected to a uniform pressure on all its faces. Knowing that the change in length of edge AB is $-3.05 \times 10^{-3} \text{ cm}$, determine (a) the change in length of the other two edges, (b) the pressure p applied to the faces of the block. (c) the change in volume of the block. Assume $E = 200 \text{ GPa}$ and $\nu = 0.29$



Let p be the pressure (compressive stress) acting on all the sides.

$$\begin{aligned}\epsilon_{xx} &= \frac{\sigma_{xx}}{E} - \nu \frac{\sigma_{yy}}{E} - \nu \frac{\sigma_{zz}}{E} = \frac{-p}{E} + \nu \frac{p}{E} + \nu \frac{p}{E} = -\frac{p}{E}(1 - 2\nu) \\ \epsilon_{yy} &= \frac{\sigma_{yy}}{E} - \nu \frac{\sigma_{xx}}{E} - \nu \frac{\sigma_{zz}}{E} = \frac{-p}{E} + \nu \frac{p}{E} + \nu \frac{p}{E} = -\frac{p}{E}(1 - 2\nu) \\ \epsilon_{zz} &= \frac{\sigma_{zz}}{E} - \nu \frac{\sigma_{xx}}{E} - \nu \frac{\sigma_{yy}}{E} = \frac{-p}{E} + \nu \frac{p}{E} + \nu \frac{p}{E} = -\frac{p}{E}(1 - 2\nu)\end{aligned}$$

$$\epsilon_{xx} = \frac{\delta_{AB}}{L_{AB}} = \frac{-3.05 \times 10^{-3} \text{ cm}}{10 \text{ cm}} = -0.305 \times 10^{-3}$$

$$\therefore -0.305 \times 10^{-3} = -\frac{p}{E}(1 - 2\nu) = -\frac{p}{200 \times 10^9}(1 - 2 \times 0.29) \quad \Leftrightarrow \quad p = 145.2 \times 10^6 \text{ Pa}$$

$$\epsilon_{yy} = \epsilon_{zz} = -\frac{p}{E}(1 - 2\nu) = -0.305 \times 10^{-3}$$

$$\delta_{BC} = L_{BC} \times \epsilon_{yy} = -1.525 \times 10^{-3} \text{ cm}$$

$$\delta_{BD} = L_{BD} \times \epsilon_{zz} = -2.2875 \times 10^{-3} \text{ cm}$$

The bulk modulus, $K = \frac{E}{3(1-2\nu)} = \frac{200 \times 10^9}{3(1-2 \times 0.29)} = 158.7 \times 10^9 \text{ Pa}$

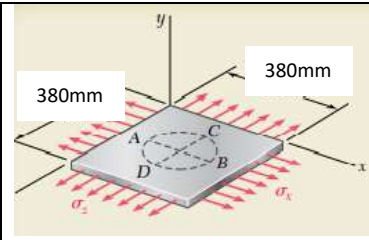
$$K = \frac{\text{hydrostatic stress}}{\text{dilatation}} = \frac{-145.2 \times 10^6}{e} \quad \Leftrightarrow \quad e = -9.1476 \times 10^{-4}$$

$$V = 10 \text{ cm} \times 7.5 \text{ cm} \times 5 \text{ cm} = 375 \text{ cm}^3$$

$$\frac{\Delta V}{V} = -9.1476 \times 10^{-4} \quad \Leftrightarrow \quad \Delta V = -0.3430 \text{ cm}^3$$

Example 2

A circle of diameter $d = 225 \text{ mm}$ is scribed on an unstressed aluminium plate of thickness $t = 20 \text{ mm}$. Forces acting in the plane of the plate later cause normal stress $\sigma_{xx} = 83 \text{ MPa}$ and $\sigma_{zz} = 138 \text{ MPa}$. For $E = 70 \text{ GPa}$ and $\nu = 0.33$, determine the change in (a) the diameter AB, and CD, (b) the thickness of the plate, (d) the volume of the plate.



The original volume of the plate, $V = 0.38\text{m} \times 0.38\text{m} \times 0.02\text{m} = 2.888 \times 10^{-3} \text{ m}^3$

	Strain in the x-direction	Strain in the y-direction	Strain in the z-direction
Stress along x-axis $\sigma_{xx} = 83 \text{ MPa}$	$\frac{\sigma_{xx}}{E} = \frac{83 \times 10^6}{70 \times 10^9} = 1.186 \times 10^{-3}$ (linear strain)	$-\nu \frac{\sigma_{xx}}{E} = -0.344 \times 10^{-3}$ (lateral strain)	$-\nu \frac{\sigma_{xx}}{E} = -0.344 \times 10^{-3}$ (lateral strain)
Stress along z-axis $\sigma_{zz} = 138 \text{ MPa}$	$-\nu \frac{\sigma_{zz}}{E} = -0.651 \times 10^{-3}$ (lateral strain)	$-\nu \frac{\sigma_{zz}}{E} = -0.651 \times 10^{-3}$ (lateral strain)	$\frac{\sigma_{zz}}{E} = \frac{138 \times 10^6}{70 \times 10^9} = 1.97 \times 10^{-3}$ (linear strain)
	$\epsilon_{xx} = 0.535 \times 10^{-3}$	$\epsilon_{yy} = -0.995 \times 10^{-3}$	$\epsilon_{zz} = 1.626 \times 10^{-3}$

Change in the diameter AB, $\delta_{AB} = \epsilon_{xx} \times (AB) = 0.535 \times 10^{-3} \times 225\text{mm} = +0.1204 \text{ mm}$

Change in the diameter CD, $\delta_{CD} = \epsilon_{zz} \times (CD) = 1.626 \times 10^{-3} \times 225\text{mm} = +0.3659 \text{ mm}$

Change in the thickness, $\delta_t = \epsilon_{yy} \times (t) = -0.995 \times 10^{-3} \times 20\text{mm} = -0.0199 \text{ mm}$

Final length of the plate, $L_x = 380(1 + 0.535 \times 10^{-3}) = 380.2033 \text{ mm}$

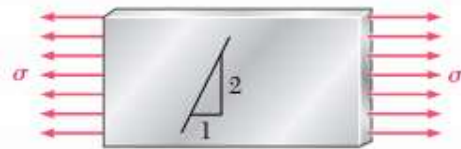
Final breadth of the plate, $L_z = 380(1 + 1.626 \times 10^{-3}) = 380.6179 \text{ mm}$

Final thickness of the plate, $L_y = 20 - 0.0199 = 19.9801 \text{ mm}$

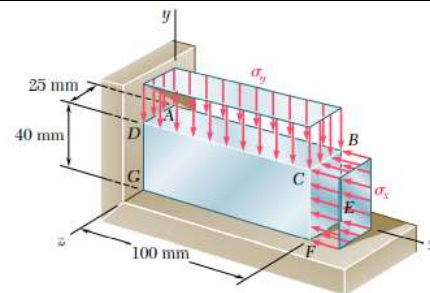
New volume of the plate, $V' = 380.2033 \text{ mm} \times 380.6179 \text{ mm} \times 19.9801 \text{ mm} = 2.891 \times 10^{-3} \text{ m}^3$

#Exercise 1

An aluminium plate ($E = 74 \text{ GPa}$, $\nu = 0.33$) is subjected to a centric axial load that causes a normal stress σ . Knowing that, before loading, a line of slope 2:1 is scribed on the plate, determine the slope of the line when $\sigma = 125 \text{ MPa}$.

**#Exercise 2**

The block shown is made of a magnesium alloy ($E = 45 \text{ GPa}$, $\nu = 0.35$). Knowing that $\sigma_x = -180 \text{ MPa}$, determine (a) the magnitude of σ_y for which the change in the height of the block will be zero, (b) the corresponding change in the area of the face ABCD, (c) the change in volume of the block.



Reference:

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3. *Engineering Mechanics of Solids*, 2/e, E. P. Popov, Pearson Education Asia.
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